Apex PdM: Video Script (4:30, YouTube-explainer style) and 10-Slide Deck

> **Competition:** Sime Darby Property University Hackathon 2026 (*Township of Tomorrow*). Rules, dates, rubric: `SDP_HACKATHON_BRIEFING_SUMMARY.md`. Roles and deadlines: `TEAM_TASK_DISTRIBUTION.md`. Step-by-step role instructions: `ROLE_GUIDES_C_D_E.md`. Clean read-aloud lines per speaker: `SCRIPT_READ_ALOUD.md`.
> **Track (official, pick one):** PLANET → **02 Circular Economy**. Energy Efficiency and People/Digital Living appear in the story only, not as extra tracks.
> **Due:** Fri 23 Oct 2026, 11:59 PM (internal target 6 PM). Deck PDF (max 10 main slides) + ~5-min video + prototype/code link.
> **Numbers:** only those in `docs/proposals/PROPOSAL_WRITING_GUIDE.md` §4 (the two DOE statistics were added there on 20 Oct, with their usage rules).
> **Team name:** Apex PdM (same as the prototype).
> **v3.1, 20 Oct 2026 (12:30):** numbers re-counted from the files after the 20 Oct wash (11 sessions, 8,379 rows); one labelled model-test line added to the demo segment and slide 8. **v3, 20 Oct 2026.** Previous version: `archive/docs_superseded_2026-10-20/VIDEO_SCRIPT_AND_SLIDE_BLUEPRINT_v2_before_youtube_style.md`.

Decisions locked on 20 Oct

Decision	Effect on this script
Full English	All lines in English
On camera is flexible (the rules do not require faces)	Every segment works as on-camera or voice-over (VO). Recommended: at least Adam Danish and Chik Amir Ashraf on camera once each, for credibility
One labelled model-test line only (changed 20 Oct, was "no model-result claims")	The leave-one-wash-out result on the 8 earlier washes may be shown, always with the words "simulated" and "eight earlier washes". No real-fault claims, no "predict". Real fault runs happen only if we reach the Top 10
No SDP or Kolaxus logo	Nowhere in the video or deck
No mosque or mall filming	Township visuals = **the compressor and water pump at home** + **royalty-free stock footage** of AC units (M01–M03, sourced by Role E) + generated graphics. The script still talks about mosques; it just doesn't film one
Adam Danish edits the video with Claude	Edit cues below are written so Claude can build them with Python + ffmpeg on Adam's PC (see Part 4)
Problem section	Kept, short (30 s), because "Problem" is a required deck area and gives the township stakes. Cut it to 15 s if time is tight

Part 1: Flow

Target **4:30** (30 s under the 5:00 limit). About 580 spoken words: ~4:00 of speech at ~145 words/min, plus the 12 s end card and short beats between segments. At a slower 130 words/min it is ~4:27 of speech, so read briskly.

Time	Segment	Speaker	Purpose
0:00–0:20	Hook	P1 Host (Role C)	Washer sound → mosque AC. Township in the first 15 seconds
0:20–0:50	Problem	P2 (Role D)	Run-to-failure; one cited statistic
0:50–1:05	"Wait, how is it done?"	P1 Host	Open loop, then "yes, it's real"
1:05–2:05	Hardware	P3 **Adam Danish**	The node, why a washer, on-device processing
2:05–3:05	Live demo + data	P4 **Chik Amir Ashraf**	Dashboard during a real cycle; data collected; model test result
3:05–4:05	Township + roadmap	P5 (Role E)	Mosque, peer comparison idea, 3-step roadmap
4:05–4:30	Close + end card	P1 + everyone available	Tagline, live link, names

Retention rules used throughout:
- **Every segment ends on a question or a tease** the next speaker answers.
- **Something changes on screen every 3–5 s:** a cut, a punch-in, a text pop or b-roll.
- **Proof before vision:** you see the real rig and live data before the township pitch.

Part 2: Script with edit cues

Legend: **VO** = voice-over allowed. **[TXT]** = on-screen text pop. **[SFX]** = sound effect. **[GFX]** = graphic Claude generates. **[CUT→ ID]** = cut to shot from the shot list (Part 3). Captions are burned in for every spoken line (not repeated below).

0:00–0:20 · HOOK · P1 (VO for first 8 s, then on camera or VO)

Time	Visual / edit	Line
0:00	Black screen. Washer spin sound fades in loud. [TXT] "Hear that?"	*(sound only, 2 s)*
0:02	[CUT→ B01] drum spinning, slow punch-in	"Hear that? That's a washing machine at full spin."
0:06	[SFX whoosh] [CUT→ M01] stock: row of outdoor AC units. [TXT] "Now imagine…"	"Now imagine that same strain inside the air-conditioning at your mosque… ten minutes before Friday prayers."
0:12	[CUT→ A01 or GFX title]	"Machines warn us before they break. We just don't listen. So we built something that does."
0:17	[GFX] title card "APEX PdM: hear it before it breaks" + [SFX hit]	"We're Team Apex PdM. And this is what we built."

0:20–0:50 · PROBLEM · P2 (on camera or VO)

Time	Visual / edit	Line
0:20	[CUT→ M02] AC units / [CUT→ H03] pump	"A township runs on machines that spin: air-conditioning, pumps, fans."
0:25	[GFX stat card] **"Over 55% of maintenance is still 'run it till it breaks'"**, source line: *US DOE FEMP, O&M Best Practices Guide, Rel. 2.0 (2004)*	"A US Department of Energy guide found over half of facility maintenance in America is still: run it till it breaks."
0:33	Quick 3-shot montage: [GFX] broken part icon → repair bill → scrap pile	"That means emergency repairs, people sweating through a prayer, and whole machines scrapped over one small part."
0:41	[GFX stat card] **"70–75% fewer breakdowns"** with predictive maintenance, same source line, plus "(industry surveys cited in the guide)"	"Yet the same guide cites surveys where predictive maintenance eliminated 70 to 75 percent of breakdowns. The tools are costly, so most machines go unwatched."
0:49	Hard cut to black, 0.3 s	*(beat)*

0:50–1:05 · "WAIT, HOW IS IT DONE?" · P1 (on camera preferred)

Time	Visual / edit	Line
0:50	[CUT→ A02] P1 to camera, punch-in on "wait". [TXT] "Wait…"	"Wait. So you might be wondering… how is it done? Is a real prototype even possible?"
0:58	[SFX riser → hit] on "Yes". [CUT→ R01] reveal shot of the pod on the washer	"Yes. And it's not a mock-up. It's running on a real washing machine, in a real, noisy home. Let's open it up."

1:05–2:05 · HARDWARE · P3 Adam Danish (on camera recommended, holding the pod)

Time	Visual / edit	Line
1:05	[CUT→ A03] Adam holding the open pod	"This is it. An ESP32 is the brain…"
1:08	[CUT→ R02–R05] macro of each part, each with a [TXT] label that pops in	"…this accelerometer feels the drum shake, this tiny microphone hears the motor, and this smart plug measures power."
1:17	[GFX] three icons: FEEL · HEAR · POWER + [TXT] **"≈ RM 170 kit"**	"Feel, hear, power. Three senses on one machine, for about 170 ringgit off the shelf."
1:24	[CUT→ B02] washer in spin, shaking	"Why a washing machine? It's a brutal test: it shakes, sloshes, and keeps changing speed."
1:31	[CUT→ H01, H02] compressor and water pump at home. [TXT] "Running right next to it"	"And our home makes it harder: an air compressor and a water pump run right beside it, the same kind of machines you'd find in a plant room."
1:40	[GFX] data-flow animation: sensors → chip → numbers → cloud → website. [TXT] **"No recordings uploaded"**	"Here's the clever part. The processing happens on the chip. Every five seconds it turns raw sound and vibration into a set of numbers, and sends only those. No sound recordings are uploaded to the cloud. That matters in homes, and in places of worship."
1:55	[CUT→ R06] pod stuck on the outside of the washer	"And it mounts on the outside. Nothing inside the machine is touched. So… does it actually work?"

2:05–3:05 · LIVE DEMO + DATA · P4 Chik Amir Ashraf (VO over screen recording; on camera for first and last line if possible)

Time	Visual / edit	Line
2:05	[CUT→ A04] Amir, then [CUT→ S01] dashboard screen recording	"Enough talking. Let's watch it work. This is our live dashboard, streaming straight from the washer."
2:12	S01 with punch-in on each card in turn	"Vibration, sound and power, updated as it runs."
2:17	[CUT→ S02] spin ramp. Split screen: dashboard left, drum (B01) right. [TXT] **"20× more vibration"**	"Now watch what happens when the drum spins up. Vibration jumps more than twenty times above idle."
2:26	Punch-in on the dominant-frequency readout. [TXT] **"19.9 Hz ≈ 1,200 rpm"**	"And the main shaking frequency climbs to almost 20 hertz. That's the drum turning about 1,200 times a minute, measured from outside the machine."
2:36	[GFX] counter animation: **"11 wash sessions · 8,379 rows of data"**	"So far: eleven recorded wash sessions, over eight thousand rows of data."
2:41	[GFX] simple diagram: many "normal" dots in a cluster; one wash hidden, then checked against the rest. Then stat card **"0 of 8 falsely flagged · 7 of 8 simulated faults caught"**, small print: "*Leave-one-wash-out test on 8 earlier washes (older firmware). Simulated fault, not a real one. Small sample.*"	"Then the AI learns this machine's normal, no broken machines needed. We tested it fairly: hide one wash, learn from the rest, check the hidden one. Across eight earlier washes, none was falsely flagged. And a simulated sound fault was caught seven times out of eight."
2:58	[CUT→ A04] Amir	"And if it works on a washer… where else could it go?"

3:05–4:05 · TOWNSHIP + ROADMAP · P5 (VO over stock footage and graphics)

Time	Visual / edit	Line
3:05	[CUT→ M01, M02] stock AC units	"Why does a washing machine matter to a township? Because it's the hardest practice run."

Time	Visual / edit	Line
3:11	[CUT→ M03] stock: large hall with AC units (or [GFX] illustrated prayer hall with AC units if no suitable clip)	"Think about a mosque. Rows of air-conditioning working hardest at Friday prayers, Ramadan nights and weddings. The same in temples, malls and clubhouses."
3:20	[GFX] AC unit with a red X, then crowd icons	"Today, when one fails, the first people to find out are the ones sitting under it."
3:26	[GFX] row of identical AC icons, each with a small pod; one turns amber. [TXT] "Spot the odd one out"	"Now picture a small pod on every unit, each learning its own normal and comparing itself with its identical neighbours. The odd one out gets flagged early."
3:37	[GFX] two cards: **PLANET**: repair, not replace · **PEOPLE**: the hall stays cool	"That's Planet: fix a small part instead of replacing the whole machine, and spot machines wasting power. And People: the prayer hall stays cool."
3:47	[GFX] 3-step roadmap appears step by step	"Our roadmap. One: a pilot on about ten AC units at one township site. Two: a custom board built for the job. Three: one dashboard watching the whole township."

4:05–4:30 · CLOSE · P1 (+ everyone available, on camera)

Time	Visual / edit	Line
4:05	[CUT→ A05] team shot (or P1 alone + team photo)	"Machines can't tell us they're tired. So we built something that listens."
4:12	[SFX hit] [GFX] logo card	"Apex PdM. Hear it before it breaks."
4:16	[GFX] end card (hold 12 s): live link **washer-pdm.pages.dev**, track "Planet: Circular Economy", 5 names + universities + roles, code link	*(music only)*

Length check (counted): 586 spoken words (v3.1). Per segment at 145 wpm: Hook 20 s, Problem 29 s, Wait 16 s, Hardware 60 s, Demo 55 s, Township 55 s, Close 8 s. Rehearse each segment with a stopwatch; the clean read-aloud lines are in `SCRIPT_READ_ALOUD.md`. If a segment runs over: trim the problem's second stat line first, then the "Why a washing machine?" line.

Part 3: Shot list (what to film and photograph)

Recording spec for everything: phone in **landscape**, 1920×1080, 30 fps, lens wiped, exposure locked (tap and hold). Shoot each shot **twice**, 3 s of stillness before and after. Name files `<ID>_take<n>.mp4` (or `.jpg`).

Voice: record lines **away from the compressor and pump** (quiet room, soft furnishings), phone 20–30 cm from the mouth, or a second phone as a voice recorder. One file per segment, e.g. `VO_P3_hardware_take1.m4a`. VO for b-roll segments can be recorded in a bedroom.

At home (Adam Danish) · by Thu 22 Oct

ID	Shot	How
B01	Washer drum spinning, through the door	Tripod or phone on a stack of books, 15 s, spin phase; also record the **sound** cleanly (phone near the door) for the hook
B02	Whole washer shaking in final spin	Wide, 10 s
R01	**Reveal**: pod on the washer	Slow push-in toward the pod, 5 s
R02	ESP32 macro	Pod open, close-up, good light
R03	MPU-6050 accelerometer macro	same
R04	INMP441 microphone macro	same
R05	Tapo P110 smart plug	Plugged in, the light on

ID	Shot	How
R06	Pod mounted on the outside of the washer, hand pointing at the tape mount	5 s
H01	Air compressor running	8 s, plus sound
H02	Water pressure pump running	8 s, plus sound
H03	Close-up of a pump or motor housing (any rotating machine)	5 s
P01–P04	**Photos**: rig full view, pod open, wiring, washer with pod	Clean background, **no addresses, family members or personal items**

Screen recordings (Chik Amir Ashraf + Adam Danish) · by Wed 21 Oct

ID	Shot	How
S01	Dashboard overview during a real wash	Full-screen browser at 1920×1080, **no bookmarks bar, no notifications**, 2–3 min
S02	The **spin-up**: Card 1 and Card 2 rising, dominant frequency climbing	Record the whole final spin (long recording; we cut it). Note the clock time when spin starts
S03	Backup: the same from a recorded session, in case the live recording misbehaves	

Stock footage (Role E) · by Wed 21 Oct

No mosque or mall filming. Download royalty-free clips from **Pexels** or **Pixabay** (check each clip's licence page; save its link in `Hackathon/video/raw/STOCK_SOURCES.txt`). Landscape, 1080p or higher. No recognisable faces, logos, brand names, signboards or watermarks.
ID	Shot	Search terms
M01	Row of outdoor AC condenser units, fans turning	"air conditioner condenser", "AC outdoor unit", "HVAC rooftop"
M02	One condenser fan spinning, close-up	"air conditioner fan", "condenser fan spinning"
M03	Large indoor hall or room with wall/ceiling AC units (a mosque interior is ideal if no faces are visible)	"mosque interior", "ceiling air conditioner", "air conditioner indoor unit"

If nothing suitable is found for M03, Claude draws it as a graphic instead.

Talking heads (any speaker who chooses on camera) · Thu 22 Oct

ID	Shot	How
A01–A05	P1 hook line, P1 "Wait…", Adam with the pod, Amir intro/outro, team or P1 close	Window light on the face, plain background, phone at eye level, framed chest-up, look at the lens. A05: everyone available in one frame (or each films their own and we combine into a grid)
T01	Group photo of all five (for the end card and the deck's team slide)	Can be taken at university

Audio and music (anyone) · by Wed 21 Oct

- **Music:** royalty-free only, downloaded from the YouTube Audio Library or Pixabay Music. Pick one upbeat electronic or lo-fi track, about 2 min long (it will loop). **Save the source link and licence text** in `Hackathon/video/music/LICENCE.txt`.
- **Sound effects:** whoosh, riser, impact hit (Pixabay sound effects). Keep the source links.

Part 4: Edit style and how the edit gets built

Style: a YouTube-explainer look in 16:9 (1920×1080, 30 fps). Energetic but not meme-heavy, because the judges are senior SDP leadership.

Element	Spec
Captions	Burned in, bottom-centre, 2 lines max, bold sans-serif, white with a dark outline; the key word in each line highlighted in the accent colour
Text pops	Big keyword or number, centre or upper third, 0.3 s scale-in, held 1.5–2.5 s
Punch-ins	Talking heads alternate 100% and 115% framing every sentence
B-roll	At most 6 s of an unbroken talking head
Colours	Match the dashboard's dark theme: near-black background, one accent colour, white text
Font	One family only (e.g. Montserrat ExtraBold for pops, Montserrat SemiBold for captions; free licence)
SFX	Whoosh on segment changes, riser + hit at "Yes", hit on the title and logo. Nothing else
Music	Low under voice (ducked), up for the end card
Lower thirds	Name + role the first time each speaker appears (e.g. "Adam Danish · Hardware & Data · UiTM"). Universities: Adam Danish and Chik Amir Ashraf UiTM, Aithish UMPSA, Manik UKM, Towa UPM

How it gets built: Claude writes `Hackathon/video/build_video.py` (Python + ffmpeg). It generates every [GFX] card and lower-third, cuts the clips to the timings above, adds punch-ins, captions, SFX and ducked music, and exports `apex_pdm_video.mp4`. The build runs on Adam's PC, where the footage is; Adam Danish runs it with Claude, owns the cut and does the upload.

- **One-time setup:** install ffmpeg (`winget install ffmpeg`) and Python 3.12.
- **Folder layout:** `Hackathon/video/raw/` (clips and photos by shot ID), `Hackathon/video/vo/` (voice files), `Hackathon/video/music/` (music, SFX, licences).
- **Review loop:** Claude builds draft 1 → the team watches it and lists fixes with timestamps → Claude builds draft 2 → final export.
- **Upload:** an **unlisted** YouTube or Google Drive link, tested in an incognito window.

Part 5: 10-slide deck (PDF, max 10 main slides)

No SDP or Kolaxus logos. Numbers only from the writing guide §4.

#	Slide	Content	Visual
1	Title	Apex PdM · "Hear it before it breaks" · Track: Planet, Circular Economy · team names + universities (UiTM, UMPSA, UKM, UPM) · live link	P01 rig photo
2	Problem	Township machines (AC, pumps, fans) run to failure; **>55% of maintenance still reactive** (US DOE FEMP, 2004); emergency repairs, discomfort in shared spaces, scrapped machines	Flow: wear → breakdown → scrap; stat card with source
3	Target user	Facility managers of mosques, temples, malls, clubhouses; maintenance contractors; residents benefit	Persona split
4	Solution	External pod (feel, hear, power) + per-machine "normal" baseline + live dashboard	R01 / P04 photo
5	Innovation	Three senses fused; on-device processing, no recordings uploaded; learns each machine's own normal, no broken-machine examples needed; about RM 170 vs costly specialist tools	Comparison table vs a single sensor / manual inspection
6	How it works	Sensors → ESP32 (one core sensing, one core Wi-Fi) → cloud database → live website → ML (tested on earlier washes; new-firmware model next)	Architecture diagram
7	Impact	Circular economy headline: repair, not replace; energy as a hypothesis with one **labelled estimate** (RM 900/yr example from the guide); predictive maintenance eliminated **70–75% of breakdowns** in surveys cited by US DOE	3 stat cards, the estimate clearly marked "estimate"
8	Feasibility + evidence	Kit about RM 170 (receipts); 11 wash sessions, 8,379 rows; 20× vibration jump; 19.9 Hz ≈ 1,200 rpm in both new-firmware washes; leave-one-wash-out test on 8 earlier washes: 0 falsely flagged, simulated sound fault caught in 7 of 8 (small sample); live dashboard; TRL 4–5	BOM table, dashboard screenshot, QR to live site

#	Slide	Content	Visual
9	Roadmap	Next: more cycles + train and test the ML; noise baselines. Then: pilot on about ten AC units at one township site. Then: custom board, industrial accelerometer, fleet dashboard	Timeline
10	Team + ask	5 members and roles; what we need: one site for a pilot	T01 photo, QR codes
Appx	Technical	Firmware architecture, feature list (32 signal features per 5.024 s window), ML design and evaluation plan, spectral plots	Charts

Note on ML evidence (updated 20 Oct): the video (2:41) and slide 8 show one labelled line from the nested leave-one-cycle-out report (`ml/models/loco_report.md`). Put the full table in the appendix, with the 95% upper bound (31%) and the limit a judge may ask about: added vibration was **not** detected in that test. The report puts the limit down to the older data (5 s snapshots, no phase labels), not the model choice. The firmware-07 model needs 3+ cycles first (2 recorded as of 20 Oct).

Part 6: Before recording and submitting

- [x] Team name: Apex PdM.
- [ ] Confirm who takes Roles C, D and E (`TEAM_TASK_DISTRIBUTION.md`); decided later.
- [x] The two DOE statistics added to `docs/proposals/PROPOSAL_WRITING_GUIDE.md` §4 (20 Oct).
- [ ] Stock clips M01–M03 downloaded, with source links saved (Role E).
- [ ] S01/S02 dashboard recording during a real wash, plus a backup recording (S03).
- [ ] All footage named by shot ID in `Hackathon/video/raw/` .
- [ ] Music and SFX licence links saved.
- [ ] Rotate the Firebase secret and Wi-Fi password and check repo history **before** sharing the code link.
- [ ] IP clause: everything submitted becomes Sime Darby Property's property. Supervisor and university check.
- [ ] Final video: 1080p MP4, under 5:00, captions burned in, unlisted link tested in incognito.
- [ ] Deck PDF: open it and check every page by eye.

Source for the problem statistics

US Department of Energy, Federal Energy Management Program. *Operations & Maintenance Best Practices: A Guide to Achieving Operational Efficiency*, Release 2.0 (PNNL-14788, July 2004), §5.2 (maintenance mix: >55% reactive) and §5.4 (predictive maintenance survey averages: 70–75% elimination of breakdowns, 25–30% reduction in maintenance costs).
https://www.pnnl.gov/main/publications/external/technical_reports/pnnl-14788.pdf
Caveat: these are US figures from a 2004 guide (the reactive share comes from a 2000 study). On screen, show the source and year; never present them as Malaysian or township-measured figures.